

Sharp Quantitative Faber–Krahn Stability via Route δ (Plan 2: a No-Selection, No-Graph, No-Schauder Proof)

Plan 2 / Route δ : assembled document

Abstract

We give a proof of the sharp quantitative Faber–Krahn inequality

$$\left(\frac{|\Omega|}{|B_1|}\right)^{2/n} (\lambda_1(\Omega) - \lambda_1(B^*)) \geq c_{\text{FK}}(n) \mathcal{A}(\Omega)^2,$$

with sharp exponent 2 on the Fraenkel asymmetry $\mathcal{A}$ (equivalently 1/2 on the scale-normalised spectral deficit) that is structurally distinct from the Brasco–De Philippis–Velichkov [11] proof and from the contradiction-free Plan 1 / Route A proof [8]. The present proof, which we call *Route δ* , makes no use of any of the following four classical ingredients: (i) the selection / penalisation principle of Cicalese–Leonardi and BDV; (ii) the graph-entry / Hausdorff-to- $C^{1,\gamma}$ step driven by free-boundary regularity; (iii) Schauder bootstrap with Gagliardo–Nirenberg interpolation; (iv) the nearly-spherical Fuglede-type closure theorem. In their place it uses a finite-perimeter level-set deficit identity, a metric foliation $\rho \mapsto [F_\rho] \in \mathcal{X}$ in the L^1 -quotient $\mathcal{X}$ of characteristic classes modulo translations, a centroid kinematic identity, a space-time radial-moment identity, and a slicing-then-squaring decomposition that converts an integrated quadratic boundary oscillation into a sharp metric trace. The proof is organised into seven self-contained building blocks (LEVELSETIDENTITY, METRICFRAMEWORK, CENTROIDBOUND, SPACETIMEIDENTITY, WEIGHTEDMETRICTRACE, BOUNDARYLAYERTRANSFER, GLOBALASSEMBLY), each delivered as a separate L^AT_EX document. The last two stages of the chain (bounded reduction and the Kohler–Jobin rearrangement) are imported verbatim from the Plan 1 Final/ library, so that the methodological novelty is concentrated in the first five blocks. The route avoids the pointwise quadratic Fusco–Julin oscillation (B^2) entirely; only its integrated form is consumed.

Contents

1	Introduction	2
1.1	The Faber–Krahn inequality	2
1.2	Stability: prior work	2
1.3	Plan 2’s contribution	3
1.4	Organisation of the paper	4
2	Notation and standing hypotheses	4
3	The seven building blocks chained	5
3.1	Block I: LEVELSETIDENTITY	5
3.2	Block II: METRICFRAMEWORK	5
3.3	Block III: CENTROIDBOUND	5
3.4	Block IV: SPACETIMEIDENTITY	5
3.5	Block V: WEIGHTEDMETRICTRACE	5
3.6	Block VI: BOUNDARYLAYERTRANSFER	6
3.7	Block VII: GLOBALASSEMBLY	6

4	The main theorem and final assembly	6
5	Constants chain	7
6	Comparison with Plan 1 (Route A)	7
7	Open issues and audit trail	8

1 Introduction

1.1 The Faber–Krahn inequality

Let $\Omega \subset \mathbb{R}^n$, $n \geq 2$, be an open set of finite Lebesgue measure and let

$$\lambda_1(\Omega) = \min_{u \in H_0^1(\Omega) \setminus \{0\}} \frac{\int_{\Omega} |\nabla u|^2 dx}{\int_{\Omega} u^2 dx}$$

be the first Dirichlet eigenvalue of $-\Delta$ on Ω . The *Faber–Krahn inequality* asserts that among open sets of a prescribed volume the ball uniquely minimises λ_1 : writing B^* for the ball with $|B^*| = |\Omega|$,

$$\lambda_1(\Omega) \geq \lambda_1(B^*), \quad (1)$$

with equality if and only if Ω is a ball (up to a set of capacity zero). This was conjectured by Lord Rayleigh in the planar case in the late nineteenth century and proved in general dimension by Faber [26] and Krahn [27] via Schwarz symmetrisation and the Pólya–Szegő principle; modern accounts appear in [28, 29, 30].

The natural way to measure the deviation from spherical symmetry is the *Fraenkel asymmetry*

$$\mathcal{A}(\Omega) = \inf_{x_0 \in \mathbb{R}^n} \frac{|\Omega \Delta (B^* + x_0)|}{|B^*|} \in [0, 2), \quad (2)$$

which vanishes precisely on translates of the ball. A quantitative Faber–Krahn inequality is an estimate of the form $(|\Omega|/|B_1|)^{2/n}(\lambda_1(\Omega) - \lambda_1(B^*)) \geq c\omega(\mathcal{A}(\Omega))$ for some modulus ω . The ellipsoidal perturbation family $\Omega_{\varepsilon} = \{|x'|^2 + (1 + \varepsilon)x_n^2 \leq 1\}$ shows $\mathcal{A}(\Omega_{\varepsilon}) \simeq \varepsilon$ and $\lambda_1(\Omega_{\varepsilon}) - \lambda_1(B^*) \simeq \varepsilon^2$, so the sharp modulus is $\mathcal{A}^2$.

1.2 Stability: prior work

The first systematic study of Faber–Krahn stability is due to Hansen–Nadirashvili [31] and Melas [32] for restricted classes of sets. Bhattacharya [33] reached every planar open set with exponent 3. Fusco–Maggi–Pratelli [13], using their sharp quantitative isoperimetric inequality [12] (see also the mass-transport approach of Figalli–Maggi–Pratelli [25]), extended this to all dimensions with exponent 4 on $\mathcal{A}$.

The sharp conjecture with exponent 2 was settled in 2015 by Brasco–De Philippis–Velichkov [11]: there exists $\sigma(n) > 0$ with $|\Omega|^{2/n}\lambda_1(\Omega) - |B^*|^{2/n}\lambda_1(B^*) \geq \sigma\mathcal{A}(\Omega)^2$. The BDV strategy has two pillars: a Kohler–Jobin [19, 18] reduction of Faber–Krahn to the analogous sharp Saint–Venant stability, then a selection principle in the spirit of Cicalese–Leonardi [15] and Acerbi–Fusco–Morini [16] combined with free-boundary regularity [35] and a nearly-spherical expansion (classical for the isoperimetric deficit [36], extended to first-eigenvalue deficit in [11, Thm. 3.3]) to produce a nearly spherical contradiction. Subsequent work of Allen–Kriventsov–Neumayer [34] addresses related spectral stability inequalities via different routes.

A constructive-constant version of the BDV theorem was given in Plan 1 [8] (*Route A*): the BDV selection / contradiction is replaced by a quantitative *single-set* selection (run once on a single near-extremal set), the regularity package is applied to that single set, and the BDV

nearly spherical theorem is invoked directly; no subsequence is taken. Plan 1 retains, however, all four of the BDV technical ingredients (selection, graph entry, Schauder, nearly-spherical closure); it removes only the contradiction.

The persistent open problem the present paper addresses is whether a proof of the sharp inequality can be given that uses *none* of the four ingredients above. The price for this is a new *finite-perimeter* infrastructure for the level-set foliation of the torsion function, which is the content of Plan 2 / Route δ .

1.3 Plan 2's contribution

Plan 2 gives a proof of the sharp quantitative Faber–Krahn inequality in which the small-asymmetry chain (S)→(G)→(I)→(N) of Plan 1 is replaced by a *metric level-set chain*. Explicitly:

- (a) *No selection principle.* No penalised functional is minimised. The original domain Ω is analysed as it stands; its volume-radius foliation $E_\rho = \{u_\Omega > t(\rho)\}$ is the structural object.
- (b) *No graph entry.* No claim that $\partial\Omega$, or that any selected $\tilde{\Omega}$, is a normal graph over a sphere of class $C^{1,\gamma}$ is made. All boundary integrals are on *reduced boundaries* ∂^*E_ρ in the De Giorgi sense.
- (c) *No Schauder interpolation.* No Kinderlehrer–Nirenberg hodograph straightening, no Lieberman oblique Schauder bootstrap, no Gagliardo–Nirenberg interpolation. The proof remains in $W_{\text{loc}}^{2,p}$ throughout.
- (d) *No nearly-spherical closure theorem.* BDV Theorem 3.3 and the Fuglede $H^{1/2}$ expansion are not invoked. The deficit-to-asymmetry conversion is purely metric.

In their place, the chain uses the following four new ingredients:

- (1) *The metric foliation.* The torsion volume-radius parametrisation $E_\rho = \{u > t(\rho)\}$ with $|E_\rho| = \omega_n \rho^n$ defines a curve $\rho \mapsto [F_\rho]$ in the metric space $(\mathcal{X}, d_{\mathcal{X}})$ of L^1 -classes modulo translations (Building Block METRICFRAMEWORK). The Ambrosio–Gigli–Savaré metric framework [17] furnishes a metric derivative $|\dot{F}_\rho|_{\mathcal{X}}$, controlled by a Reshetnyak / coarea limit of the moving-region first variation.
- (2) *The centroid kinematic identity.* The bulk centroid $\bar{z}_\rho = |E_\rho|^{-1} \int_{E_\rho} x \, dx$ satisfies

$$\bar{z}'_\rho = \frac{1}{|E_\rho|} \int_{\partial^*E_\rho} (x - \bar{z}_\rho) V_\rho \, d\mathcal{H}^{n-1},$$

with $V_\rho = -t'(\rho)/|\nabla u|$ the normal velocity. The right-hand side has *no homothety term*, so the bound on $\bar{z}'_\rho$ goes through the unconditional pair (Cauchy–Schwarz on V_ρ , FMP / Fusco–Julin at the centroid) and does not consume any per- ρ radial-trace input (Building Block CENTROIDBOUND).

- (3) *The space-time Π identity.* The signed radial moment $\Pi(E_\rho, \bar{z}_\rho) = (n-1) \int (\mathbf{1}_{B_\rho(\bar{z}_\rho)} - \mathbf{1}_{E_\rho}) |x - \bar{z}_\rho|^{-1} dx$ satisfies, by Fubini in (x, ρ) , $\int_{\rho_*}^{\rho_\delta} \Pi(E_\rho, \bar{z}_\rho) \, d\mu(\rho) = (n-1) \int_{\tilde{\Omega}} \sigma(x) dx$ for an explicit Fubini density σ (Building Block SPACETIMEIDENTITY).
- (4) *Slicing-then-squaring.* The radial L^1 trace $T_1 = \int_{\partial^*E_\rho} |x - \bar{z}_\rho|/\rho - 1| \, d\mathcal{H}^{n-1}$ is split as $T_1 = T_1^{\text{rad, slic}} + T_1^{\text{tan}}$; the radial piece is bounded *linearly* via the good-ray equality of slice-wise volumes and a bad-ray multiplicity bound; the squared volume-Fraenkel $|E_\rho \Delta B_\rho|^2$ is controlled by $\Pi(E_\rho, \bar{z}_\rho)$ via a slice-rearrangement. Squaring before integrating preserves the sharp δ -rate that the alternative T_2 -route would leak at $\sqrt{\delta}$.

Theorem 1.1 (Sharp quantitative Faber–Krahn, Plan 2 form). *Let $n \geq 2$. There exists a constant $c_{\text{FK}}(n) > 0$, depending only on the dimension, given explicitly by (8) below as a composition of named constants of the seven Plan 2 building blocks (and of the Plan 1 BOUNDEDREDUCTION and KOHLERJOBINTTRANSFER blocks at the dimensional radius $R_*(n)$), such that*

$$\left(\frac{|\Omega|}{|B_1|}\right)^{2/n} (\lambda_1(\Omega) - \lambda_1(B^*)) \geq c_{\text{FK}}(n) \mathcal{A}(\Omega)^2$$

for every open $\Omega \subset \mathbb{R}^n$ of finite positive measure. The exponent 2 on $\mathcal{A}$ is sharp.

1.4 Organisation of the paper

The argument is organised as a chain of seven self-contained building blocks. Schematically,

LEVELSETIDENTITY $\rightarrow$ METRICFRAMEWORK $\rightarrow$ CENTROIDBOUND $\rightarrow$ SPACETIMEIDENTITY $\rightarrow$ WEIGHTEDM

The first six blocks are the geometric heart of Plan 2; the seventh imports the bounded reduction of [11, §5] (equivalently Plan 1’s BOUNDEDREDUCTION [9]) and the Kohler–Jobin transfer of Brasco [18] (Plan 1’s KOHLERJOBINTTRANSFER [10]).

2 Notation and standing hypotheses

Fix $n \geq 2$ and $R \geq 2$. We use the following normalisations:

- $u_\Omega \in H_0^1(\Omega)$ is the variational torsion function: $-\Delta u_\Omega = 1$ in Ω , $u_\Omega = 0$ on $\partial\Omega$.
- $T(\Omega) = \int_\Omega u_\Omega$ is the torsional rigidity, and $E(\Omega) = -T(\Omega)/2$ is the Saint–Venant energy.
- $\delta_T(\Omega) := E(\Omega) - E(B^*) \geq 0$ at fixed volume.
- $\lambda_1(\Omega)$ is the first Dirichlet eigenvalue of $-\Delta$.
- $\mathcal{A}(\Omega) = \inf_{x_0} |\Omega \Delta(B^* + x_0)|/|B^*| \in [0, 2)$.
- For $\rho \in [\rho_*, 1]$, $E_\rho = \{u_\Omega > t(\rho)\}$ with $|E_\rho| = \omega_n \rho^n$ and $\mu(d\rho) = (-t'(\rho))d\rho$.
- $\bar{z}_\rho = |E_\rho|^{-1} \int_{E_\rho} x dx$ is the centroid of E_ρ .
- $F_\rho = \rho^{-1} E_\rho$ is the rescaled level set; $[F_\rho] \in \mathcal{X}$ its class in $\mathcal{X} = \{L^1\text{-classes}\}/\mathbb{R}^n$ with $d_{\mathcal{X}}([A], [C]) = \inf_a |A \Delta(C + a)|$.
- $V_\rho(x) = -t'(\rho)/|\nabla u(x)|$, $H_{z,\rho}(x) = (x - z) \cdot \nu_{E_\rho}(x)/\rho$.
- $\rho_\delta := 1 - \kappa \sqrt{\delta_T(\Omega)}$ the deficit endpoint; $\hat{\rho} := \rho_\delta$ the metric-trace radius (upper endpoint of the order-one window $J^* = [\rho_\delta - L^*, \rho_\delta]$ where the uniform pointwise bound $d_{\mathcal{X}}(F_\rho, B_1)^2 \leq C_* \delta_T$ holds).
- $\omega_n = |B_1|$, $L_n = \lambda_1(B_1)$, $T_n = T(B_1) = \omega_n/(n(n+2))$.
- $R_*(n) = \max\{d(n), 2\}$ the BDV bounded-reduction radius; we fix $\rho_* = 1/2$.

Notation convention (two parametrisations). We use both notations: $E_t = \{u > t\}$ (level-set parametrisation, Block I) and $E_\rho = \{u > t(\rho)\}$ with $|E_\rho| = \omega_n \rho^n$ (volume-radius parametrisation, Blocks II–VII). The map $t \leftrightarrow \rho$ is the inverse distribution function.

3 The seven building blocks chained

3.1 Block I: LevelSetIdentity

Theorem 3.1 (Exact deficit identity; [1]). *Let $\Omega \subset \mathbb{R}^n$ be open with $|\Omega| < \infty$. With $E_t = \{u > t\}$, $m(t) = |E_t|$, $\alpha(t) = \int_{\partial^* E_t} |\nabla u|^{-1} d\mathcal{H}^{n-1}$, $\beta(t) = \int_{\partial^* E_t} |\nabla u| d\mathcal{H}^{n-1}$, $D_H(t) = \alpha\beta - P(E_t)^2$, $D_I(t) = P(E_t)^2 - n^2 \omega_n^{2/n} m(t)^{2-2/n}$,*

$$\boxed{\frac{2}{|\Omega|} (E(\Omega) - E(B)) = \frac{1}{|\Omega|} \int_0^{\|u\|_\infty} \frac{D_H(t) + D_I(t)}{n^2 \omega_n^{2/n} m(t)^{1-2/n}} dt.} \quad (3)$$

Both $D_H, D_I \geq 0$ a.e.; no boundedness or smoothness of Ω is needed. All integrals on the reduced boundary $\partial^ E_t$.*

3.2 Block II: MetricFramework

Theorem 3.2 (Metric first variation; [2]). *For $\Omega \subset B_R$ with $|\Omega| = \omega_n$ and any Borel centre field $\{z_\rho\}$ with $|z_\rho| \leq R'$, there is $C_{n,\rho_*,R} > 0$ such that*

$$\boxed{|\dot{F}_\rho|_{\mathcal{X}} \leq C_{n,\rho_*,R} \inf_{a \in \mathbb{R}^n} \int_{\partial^* E_\rho} |V_\rho - H_{z_\rho,\rho} - a \cdot \nu_\rho| d\mathcal{H}^{n-1}} \quad (4)$$

for a.e. $\rho \in [\rho_, 1]$.*

3.3 Block III: CentroidBound

Theorem 3.3 (H^1 -in- ρ centroid bound (F); [3]). *$\rho \mapsto \bar{z}_\rho$ is Lipschitz on $[\rho_*, 1]$ and there is $C_{\text{cent}}(n, R, \rho_*) > 0$ with*

$$\boxed{\int_{\rho_*}^{\rho_\delta} |\bar{z}'_\rho|^2 (-t'(\rho)) d\rho \leq C_{\text{cent}} \delta_T(\Omega).} \quad (\text{F})$$

The proof uses the centroid kinematic identity $\bar{z}'_\rho = |E_\rho|^{-1} \int_{\partial^ E_\rho} (x - \bar{z}_\rho) V_\rho d\mathcal{H}^{n-1}$ (no homothety term), the unconditional D_H -Cauchy on V_ρ , FMP / Fusco–Julin at the centroid, and the integrated Talenti gap of Theorem 3.1.*

3.4 Block IV: SpaceTimeIdentity

Theorem 3.4 (Space-time Π identity and integrated bounds; [4]). *Conditional on (F):*

$$\boxed{\int_{\rho_*}^{\rho_\delta} \Pi(E_\rho, \bar{z}_\rho) d\mu(\rho) = (n-1) \int_{\tilde{\Omega}} \sigma(x) dx,} \quad (5)$$

the integrated quadratic oscillation $\iint |\nu - e_r|^2 d\mathcal{H}^{n-1} d\mu \leq C_{B^2} \delta_T$ (B^2 -int), and the integrated radial trace $\int (\int |H_{\bar{z}_\rho,\rho} - 1| d\mathcal{H}^{n-1})^2 d\mu \leq C_{\text{radial}} \delta_T$ (T_1^2 -int). The proof uses $T_1 = T_1^{\text{rad,slic}} + T_1^{\text{tan}}$ with linear good-ray equality and the slice-rearrangement $|E_\rho \Delta B_\rho|^2 \leq C \Pi(E_\rho, \bar{z}_\rho)$.

3.5 Block V: WeightedMetricTrace

Theorem 3.5 (Weighted metric trace at ρ_δ ; [5]). *Conditional on (F), $\int (d_{\mathcal{X}}(F_\rho, B_1)^2 + |\dot{F}_\rho|_{\mathcal{X}}^2) d\mu \leq C \delta_T$ (integrated action, encoding $\int D_I d\mu$ via Fusco–Julin and $\int D_H d\mu$ via the metric first variation), $\int |\dot{F}_\rho|_{\mathcal{X}}^2 d\rho \leq C' \delta_T$ (kinetic bound). Combining the two on the order-one window*

$J^* = [\rho_\delta - L^*, \rho_\delta]$ with $L^* = (\rho_\delta - \rho_*)/4$ via Markov on the distance term plus kinetic Cauchy–Schwarz transport gives the uniform pointwise bound

$$\sup_{\rho \in J^*} d_{\mathcal{X}}(F_\rho, B_1)^2 \leq C_* \delta_T(\Omega). \quad (6)$$

In particular, setting $\hat{\rho} := \rho_\delta$, $d_{\mathcal{X}}(F_{\rho_\delta}, B_1)^2 \leq C_* \delta_T(\Omega)$ at the upper endpoint of the deficit window. Either input alone forces a $\sqrt{\delta_T}$ exponent loss; the D_H/D_I pairing yields the sharp bound at ρ_δ .

3.6 Block VI: BoundaryLayerTransfer

Theorem 3.6 (Bounded sharp Saint–Venant; [6]). *There is $C(n, R, \rho_*) > 0$ such that every open $\Omega \subset B_R$ with $|\Omega| = \omega_n$ satisfies*

$$\mathcal{A}(\Omega)^2 \leq C(n, R, \rho_*) \delta_T(\Omega). \quad (7)$$

The proof combines the endpoint case $\hat{\rho} = \rho_\delta$ of (6) with Lemma 8.3 of [6] ($\mathcal{A}(\Omega) \leq \mathcal{A}(E_{\rho_\delta}) + 2|\Omega \setminus E_{\rho_\delta}|/\omega_n$, where $|\Omega \setminus E_{\rho_\delta}| \leq \omega_n n \kappa \sqrt{\delta_T}$ since $\rho_\delta = 1 - \kappa \sqrt{\delta_T}$) and the squaring identity $(a + b)^2 \leq 2(a^2 + b^2)$; sharp exponent 2.

3.7 Block VII: GlobalAssembly

Theorem 3.7 (Sharp Faber–Krahn; [7]). *Combining (7) at the dimensional radius $R_*(n)$ with the BDV bounded reduction [9] and the Kohler–Jobin transfer [10], there is a dimensional constant $c_{\text{FK}}(n) > 0$ given by*

$$c_{\text{FK}}(n) := \min \left\{ \frac{4L_n c_{\text{SV}}^{\text{glob}}(n)}{(n+2)T_n}, \frac{L_n(2^{2/(n+2)} - 1)}{4} \right\}, \quad (8)$$

such that for every open $\Omega \subset \mathbb{R}^n$ with $0 < |\Omega| < \infty$,

$$\left(\frac{|\Omega|}{|B_1|} \right)^{2/n} (\lambda_1(\Omega) - \lambda_1(B^*)) \geq c_{\text{FK}}(n) \mathcal{A}(\Omega)^2. \quad (9)$$

4 The main theorem and final assembly

Theorem 4.1 (Sharp quantitative Faber–Krahn stability, Plan 2 form). *Let $n \geq 2$. There is a dimensional constant $c_{\text{FK}}(n) > 0$, given by (8), such that for every open $\Omega \subset \mathbb{R}^n$ of finite positive measure, $(|\Omega|/|B_1|)^{2/n} (\lambda_1(\Omega) - \lambda_1(B^*)) \geq c_{\text{FK}}(n) \mathcal{A}(\Omega)^2$. The exponent 2 on $\mathcal{A}$ is sharp.*

Proof. By scale invariance of both sides — $(|\Omega|/|B_1|)^{2/n} (\lambda_1(\Omega) - \lambda_1(B^*))$ is dimensionless, as is $\mathcal{A}(\Omega)$ — WLOG $|\Omega| = \omega_n$ after rescaling $\Omega \mapsto (\omega_n/|\Omega|)^{1/n} \Omega$. Fix $\rho_* = 1/2$. The chain Theorems 3.1–3.6 delivers the bounded Saint–Venant inequality at $R = R_*(n)$. Theorem 3.7 then internalises the R -dependence via the BDV bounded reduction and converts to Faber–Krahn via Kohler–Jobin. $\square$

5 Constants chain

Stage	Constant	Dependence
(L) deficit identity	—	—
(M) first variation	$C_{n,\rho_*,R} = \rho_*^{-n}$	n, ρ_*, R
(F) centroid bound	C_{cent}	n, R, ρ_*
(II) (B ² -int)	C_{B^2}	n, R, ρ_*
(II) (T ₁ ² -int)	C_{radial}	n, R, ρ_*
(K) action, kinetic, trace	C, C', C_*	n, R, ρ_*
(∂) bounded SV	$C(n, R, \rho_*)$	n, R, ρ_*
$\rho_* = 1/2, R \rightarrow R_*(n)$	$C(n, R_*(n), 1/2)$	n
(FK) BDV constants	$C_9(n), \delta(n), d(n)$	n
(FK) global SV	$c_{\text{SV}}^{\text{glob}}(n)$	n
(FK) KJ multiplier	$4L_n / ((n+2)T_n)$	n
(FK) final	$c_{\text{FK}}(n)$	n

Conclusion. All R - and ρ_* -dependence is absorbed at the Block (∂) $\rightarrow$ (FK) interface; final constant $c_{\text{FK}}(n)$ depends only on n .

6 Comparison with Plan 1 (Route A)

Both routes deliver Theorem 4.1. The structural difference is concentrated in the bounded sharp Saint–Venant stage. Plan 1 uses the chain (S) selection $\rightarrow$ (G) graph entry $\rightarrow$ (I) Schauder interpolation $\rightarrow$ (N) nearly-spherical closure ([8]). Plan 2 uses the chain (L) deficit identity $\rightarrow$ (M) metric first variation $\rightarrow$ (F) centroid bound $\rightarrow$ (II) space-time + slicing-then-squaring $\rightarrow$ (K) metric trace $\rightarrow$ (∂) boundary-layer transfer.

Tool	Plan 1 / Route A	Plan 2 / Route δ
Selection / penalised functional	uses	avoids
Graph entry ($C^{1,\gamma}$ normal graph)	uses	avoids
Schauder bootstrap (Lieberman)	uses	avoids
Gagliardo–Nirenberg interpolation	uses	avoids
Nearly-spherical closure (BDV Thm. 3.3)	uses	avoids
Fuglede $H^{1/2}$ expansion	uses	avoids
Free-boundary regularity (Alt–Caffarelli)	uses	avoids
BV coarea on $\partial^* E_t$ for a.e. t	not used	uses
AGS metric framework, $ \dot{F}_\rho _{\mathcal{X}}$	not used	uses
Centroid kinematic identity (no homothety)	not used	uses
Space–time II identity	not used	uses
Slicing-then-squaring (good rays / bad rays)	not used	uses
Slice-rearrangement $ E\Delta B ^2 \leq C\Pi$	not used	uses
Sobolev–Sard (Figalli–Maggi App. A [24])	not used	uses
BDV bounded reduction [11, §5]	uses	uses
Kohler–Jobin transfer	uses	uses

The methodological gain of Plan 2 is that no PDE regularity beyond $W_{\text{loc}}^{2,p}$, no free-boundary regularity, no selection, no graph parametrisation, and no nearly-spherical closure is used. The price is the new infrastructure in Blocks (M)–(II)–(K).

7 Open issues and audit trail

The Plan 2 chain has been audited in three waves on the markdown source side; each post-audit correction is reflected in the `WIP_X.tex` files.

Wave 1¹ flagged five issues: centre-choice ambiguity (Fraenkel vs. centroid); reliance on the open pointwise per- ρ radial trace (R); missing Sobolev–Sard citation; broken Cauchy–Schwarz in (G4) §5.2; notational drift between centre choices.

Wave 2² addressed: the centroid choice $\bar{z}_\rho = |E_\rho|^{-1} \int_{E_\rho} x$; the integrated form (B²-int) replacing pointwise (R); the Figalli–Maggi App. A Sobolev–Sard citation; the kinetic bound (K) rederivation from (B²-int).

Wave 3³ introduced two new structural inputs: (F) the H^1 -in- ρ centroid bound (CENTROIDBOUND); and the slicing-then-squaring decomposition (SPACETIMEIDENTITY §5). The latter replaces the broken Cauchy–Schwarz of (G4) §5.2 by the linear good-ray equality and the slice-rearrangement; squaring before integrating preserves the sharp δ rate.

Status. The Plan 2 proof closes in its integrated form only. Pointwise per- ρ statements (R) and (B²) are not invoked. The integrated form, controlled by (F) of CENTROIDBOUND and (B²-int) of SPACETIMEIDENTITY, suffices for the metric-trace and boundary-layer-transfer chain. This is the unconditional honest content of Plan 2.

References

- [1] Plan 2 building block LEVELSETIDENTITY, `WIP_LevelSetIdentity.tex`.
- [2] Plan 2 building block METRICFRAMEWORK, `WIP_MetricFramework.tex`.
- [3] Plan 2 building block CENTROIDBOUND, `WIP_CentroidBound.tex`.
- [4] Plan 2 building block SPACETIMEIDENTITY, `WIP_SpaceTimeIdentity.tex`.
- [5] Plan 2 building block WEIGHTEDMETRICTRACE, `WIP_WeightedMetricTrace.tex`.
- [6] Plan 2 building block BOUNDARYLAYERTRANSFER, `WIP_BoundaryLayerTransfer.tex`.
- [7] Plan 2 building block GLOBALASSEMBLY, `WIP_GlobalAssembly.tex`.
- [8] Plan 1 master document (companion route), `Final/master.tex`.
- [9] Plan 1 building block BOUNDEDREDUCTION, `Final/BoundedReduction.tex`.
- [10] Plan 1 building block KOHLERJOBINTRANSFER, `Final/KohlerJobinTransfer.tex`.
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¹Internal audit trail: source file `agent6-adversarial-audit.md`.

²Internal audit trail: source files `wave2-*`.

³Internal audit trail: source files `wave3-F,G,H,J`.

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